

Qatar-1b

R-2814

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250804 · created 2026-09-05T01:45:42+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460891.7839 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.017
Transit depth (Rp/Rs)^2	3.35 +/- 0.63 [%]
Orbital Inclination (inc)	83.14 +/- 1.03
Airmass coefficient 1 (a1)	1.847 +/- 0.024
Airmass coefficient 2 (a2)	-0.49 +/- 0.16
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	None
Optimal Aperture	3.59
Optimal Annulus	12.25
Transit Duration (day)	0.0657 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.183 $\pm$ 0.017 vs 0.1463 (deviation 2.16 σ)
Duration fit vs published	0.0657 d vs 0.0692 d (deviation 0.52 σ)
Mid-time O-C	0.9 min
Depth SNR	10.8
Residual scatter	1.29 %

Light curve

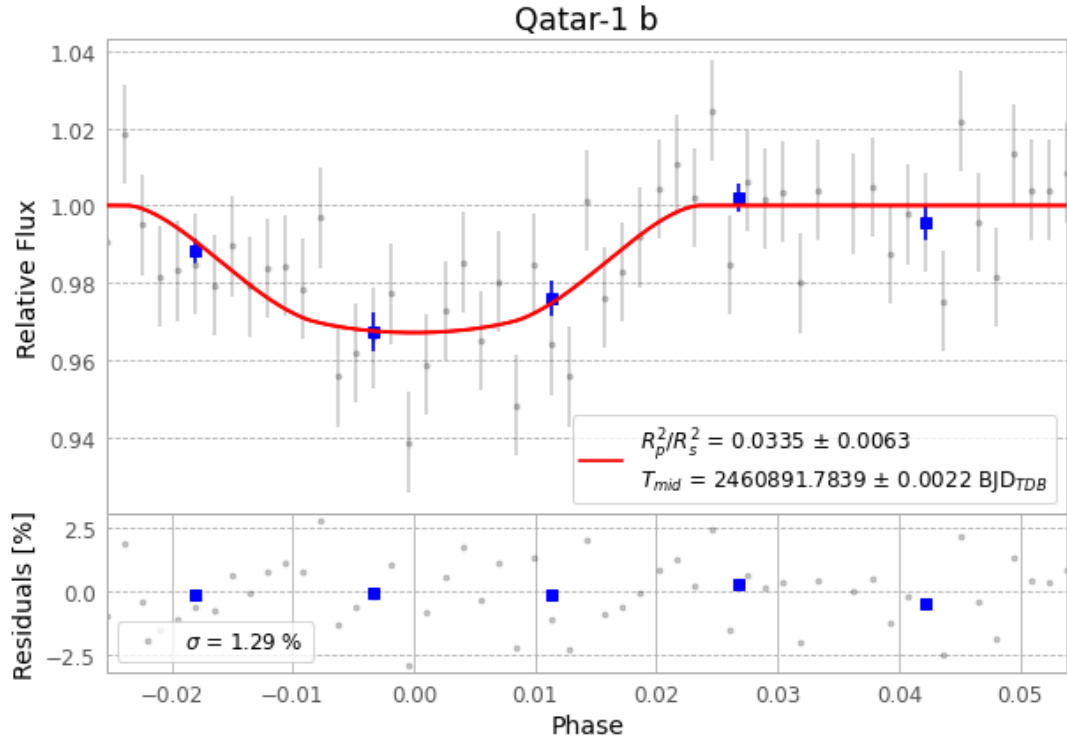


Figure 1 — FinalLightCurve_Qatar-1 b_2025-08-03.png (SHA-256 17bfef54924b38c9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e15656bf9843fc6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), Qatar-1250804055416.FITS → Qatar-1250804083615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250804.log (7693c4678cd6...), FinalLightCurve_Qatar-1 b_2025-08-03.png (17bfef54924b...), FinalLightCurve_Qatar-1 b_2025-08-03.csv (ca09e0bb7b8f...)

Compute resources

Wall time 691.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:45Z.